Tate's Thesis

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Contents

1 Local Functional Equation

To treat the local field of number fields, we fix the notation for sake of convience.

Definition 1.1. Let F be a field, then we call F is a local field if it is one of the following types:

1. Archimedean local field: $\mathbb{R}$ or $\mathbb{C}$;
2. Non-archimedean local field: a finite extension of $\mathbb{Q}_p$

Equivalently, a local field is a completion of a number field at a place, and it is non-archimedean if the place is finite.

In the case of non-archimedean local field, we use the following notation:

$$\begin{aligned} \mathcal{O} &: \text{ the valuation ring} \\ \mathcal{O}^\times &: \text{ the unit group of } \mathcal{O} \\ \mathfrak{p} &: \text{ the maximal ideal of } \mathcal{O} \\ \varpi &: \text{ a uniformizer} \\ \mathfrak{d} &: \text{ the different of } F/\mathbb{Q}_p \end{aligned}$$

Definition 1.2. Let F be a local field, then $f : F \to \mathbb{C}$ is called a Schwartz-Bruhat function if it is one of the following cases:

- When F is archimedean, f is a Schwartz function, which is a smooth and rapidly deacrasing function of class C^∞.
- When F is non-archimedean, f is locally constant and compactly supported.

Let $\mathcal{S}(F)$ denotes the $\mathbb{C}$-vector space of all Schwartz-Bruhat functions on F.

1.1 Additive Character

Let F be a local field, then in each case we can know that $(F, +)$ is a locally compact abelian group, which induces a Haar measure μ on F, then for any non-zero element $a \in F$, we can define a new measure on F by pull back:

$$\forall E \subset F \text{ measurable }, \quad \mu_a(E) = \mu(aE)$$

and we can conculde the following result:

Lemma 1.1. μ_a is a Haar measure on F, and there exists a constant factor $|a|$ such that $\mu_a = |a|\mu$, such a factor is independent of the choice of μ with

$$|a| = \begin{cases} \text{the standard absolute value if } F = \mathbb{R} \\ \text{the square of standard absolute value if } F = \mathbb{C} \\ N(\mathfrak{p})^{-v_{\mathfrak{p}}(a)} \text{ if } F \text{ is non-archimedean} \end{cases}$$

Proof. $\square$

The constant factor can be extended to the whole field F by setting $|0| = 0$, and we call it **(normalized) module** on F, it is equivalent to standarad absolute value in each case topologically, to avoid confusion, we will use the notation $|\cdot|$ to denote the module on F in the following.

Definition 1.3. Additive character of a local field F is a continous group homomorphism $\psi : (F, +) \to \mathbb{S}^1$, and $\hat{F}$ is the set of all additive characters of F, it is called the dual group of F.

The dual group is indeed a group under pointwise multiplication, for topology on $\hat{F}$, we consider the compact-open topology, i.e. the topology generated by

$$[K, U] = \{\chi \in \hat{F} : \chi(K) \subset U\}$$

for every compact subset K of F and every open subset U of $\mathbb{S}^1$. It is a natural choice for functiona space, so dual group is ineed a topological group, a remarkable result here is that local field is self-dual.

Theorem 1.2. Let ψ be a non-trival additive character of F, then the map

$$\Psi : F \to \hat{F}, \quad a \mapsto \psi_a : x \mapsto \psi(ax)$$

is an isomorphism of LCA groups.

Proof. □

Remark. A LCA group is not necessary self-dual, for example the dual group of $\mathbb{Z}$ is $\mathbb{S}^1$, and the dual group of $\mathbb{S}^1$ is $\mathbb{Z}$.

By the theory of harmonic analysis on LCA, if LCA group G is self-dual, then it is nice to identify the fourier tranform of a function f on G as a function on G itself, so it motivates us to define the transform on local field as follows:

Definition 1.4. Fix a non-trival additive character ψ and a Haar measure μ on F, the Fourier transform of type (ψ, μ) of a function $f \in L^1(F)$ is defined as

$$\hat{f}(y) = \int_F f(x)\psi(xy)\, d\mu(x)$$

There is no confusion to identify y as ψ_y by isomorphism, and the norm of ψ is always 1, so $|\hat{f}(y)| \leq \|f\|_{L^1} < \infty$ ensures that the transform is well-defined.

Definition 1.5. The standard additive character of F is defined as following:

- If $F = \mathbb{R}$, let $\psi(x) = e^{-2\pi i x}$;
- If $F = \mathbb{Q}_p$, let $\psi(x) = e^{2\pi i \{x\}}$, where $\{x\}$ is the fractional part of x defined by

$$x = \sum_{n \geq N} a_n p^n \implies \{x\} = \sum_{n=N}^{-1} a_n p^n$$

  where $\{x\} = 0$ if $N \geq 0$, i.e. $x \in \mathbb{Z}_p$.

- Let $F_0 = \mathbb{R}$ or $\mathbb{Q}_p$, and F be a finite extension of F_0, let the standard additive character of F be the compistion
$$F \xrightarrow{\mathrm{Tr}_{F/F_0}} F_0 \xrightarrow{\psi_0} \mathbb{S}^1$$
where ψ_0 is the standard additive character of F_0.

In particular, if $F = \mathbb{Q}$, the standard character is $\psi(z) = e^{-2\pi i(z+\bar{z})}$. In the case of $F = \mathbb{Q}_p$, we have $\{x\} \in \mathbb{Z}[1/p]$ and $x - \{x\} \in \mathbb{Z}_p$, the intersection $\mathbb{Z}_p \cap \mathbb{Z}[1/p] = \mathbb{Z}$ implies that standard character ψ is a non-trival homomoprhism. In the aspect of continuity, we notice that $\mathbb{Z}_p \subset \ker \psi$ as an open subset containing 0, which shows that ψ is continous at 0, then ψ is continous on $\mathbb{Q}_p$ since it is a topological group. Hence ψ is exactly a non-trival additive character of $\mathbb{Q}_p$, It is not clear than the case $F = \mathbb{R}$.

From now on, we will always use standard character in the Fourier transform, so the fourier transform will depend only on the choice of Haar measure, so a normalization can be done to obtain the inversion formula.

Proposition 1.3. Let F be a local field, then there exists a unique self-dual Haar measure μ, that means if $f \in \mathcal{S}(F)$, then $\hat{f} \in \mathcal{S}(F)$ and $\widehat{\hat{f}}(x) = f(-x)$, explicitly measure is given by following:

$$dx = \begin{cases} \text{lebesgue measure} & \text{If } F = \mathbb{R} \\ 2 \times \text{lebesgue measure if} & \text{If } F = \mathbb{C} \\ \text{the Haar measure such that } \mu(\mathcal{O}) = N(\mathfrak{d})^{-\frac{1}{2}} & \text{If } F \text{ is non-archimedean} \end{cases}$$

Proof. □

1.2 Multiplicative Character

The module on local field F induced a continous homomorphism

$$F^\times \to \mathbb{R}_{>0}, \quad x \mapsto |x|$$

the kernel of this homomorphism will be important in the following discussion, and we denote it by U, and we can conclude

$$U = \begin{cases} \{\pm 1\} & \text{If } F = \mathbb{R} \\ \mathbb{S}^1 & \text{If } F = \mathbb{C} \\ \mathcal{O}^\times & \text{If } F \text{ is non-archimedean} \end{cases}$$

The group U as a subspace of local field F is compact in each case, and also open if the F is non-archimedean.

Definition 1.6. a (multiplicative) quasi-character is a continous group homomorphism $\chi : F^\times \to \mathbb{C}^\times$, and the set of all quasi-characcters is denoted by $\mathcal{X}(F^\times)$.

a quasi-character χ is called a (unitary) character if $\mathrm{im}(\chi) \subset \mathbb{S}^1$; a quasi-character χ is called unramified if χ is trival on U.

The vocabulary here follows Tate's thesis, here are the examples in the case of archimedean local field, which is the classical result in harmonic analysis.

Example 1.4. Every quasi-character of $\mathbb{R}^\times$ is of the form

$$\chi_{\epsilon,s}(t) = \mathrm{sign}(t)^\epsilon \cdot |t|^s, \quad \epsilon \in \{0,1\}, s \in \mathbb{C}$$

Every quasi-character of $\mathbb{C}^\times$ is of the form

$$\chi_{n,s}(z) = (\frac{z}{\|z\|})^n \cdot |z|^s, \quad n \in \mathbb{Z}, s \in \mathbb{C}$$

where $\|z\|$ is the standard absolute value on $\mathbb{C}$, and in both case, the character is unitary if and only if s is a purely imaginary number.

By the example, it is natural to separate the quasi-character into two parts, the first part is a unitary character on units, and the second part is a power of module, in fact it is unramified and we can generalize the result.

Lemma 1.5. A quasi-character $\chi \in \mathcal{X}(F^\times)$ is unramified if and only if it is of the form $\chi(x) = |x|^s$ for some $s \in \mathbb{C}$. Let $\mathcal{U}(F^\times)$ denote the set of all unramified quasi-characters on $F^\times$, then it forms a subgroup of $\mathcal{X}(F^\times)$ with isomorphism

$$\mathcal{U}(F^\times) \cong \begin{cases} \mathbb{C} & \text{If } F \text{ is archimedean} \\ \mathbb{C}/\frac{2\pi i}{\log N(\mathfrak{p})}\mathbb{Z} & \text{If } F \text{ is non-archimedean} \end{cases}$$

Proof. Notice that χ is unramified if and only if χ factors through $|F^\times|$, because $U \subset \ker \chi$ if χ is trivial on U, while U is the kernel of the surjection $x \mapsto |x|$, that means there exists a continous homomorphism $\tilde{\chi} : |F^\times| \to \mathbb{C}^\times$ such that the following diagram commutes:

$$\begin{tikzcd} F^\times \arrow[r, "\chi"] \arrow[d, "x\mapsto|x|"'] & \mathbb{C}^\times \\ |F^\times| \arrow[ur, "\tilde{\chi}"'] & \end{tikzcd}$$

Hence if χ is unramified, it is enough to prove that continous homomorphism $\tilde{\chi} : |F^\times| \to \mathbb{C}^\times$ is of the form $x \mapsto x^s$ for some $s \in \mathbb{C}$. If F is archimedean, it is immediate by the example above, if F is non-archimedean, $|F^\times| = q^{\mathbb{Z}}$ with $q = N(\mathfrak{p})$, so the morphism is determined by the image of q, if we set $\tilde{\chi}(q) = a$, then we can choose $s = \log a$ with respect to a branch of logarithm.

Conversely, it is not hard to check that $\chi_s(x) = |x|^s$ is unramified for any $s \in \mathbb{C}$, which proves the first part of the lemma and induces a surjective map

$$\Phi : \mathbb{C} \to \mathcal{U}(F^\times), \quad s \mapsto \chi_s$$

and it is not hard to check $\mathcal{U}(F^\times)$ is indeed a subgroup of $\mathcal{X}(F^\times)$ and $\chi_{s+z} = \chi_s \cdot \chi_z$ for any $s, z \in \mathbb{C}$, which implies that Φ is a group homomorphism with kernel $\{s \in \mathbb{C} : |x|^s = 1, \forall x \in F^\times\}$. If F is archimedean, the kernel is trivial so Φ is an isomorphism; if F is non-archimedean, the kernel is exactly $\{s \in \mathbb{C} : |\varpi|^s = 1\} = \frac{2\pi i}{\log q}\mathbb{Z}$, so it induces a desired isomorphism. $\square$

Lemma 1.6. The map $\chi \mapsto (\chi|_U, \chi \cdot (\chi|_U \circ \rho)^{-1})$ gives an isomorphism

$$\mathcal{X}(F^\times) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{cont}}(U, \mathbb{S}^1) \times \mathcal{U}(F^\times)$$

where $\rho : F^\times \to U$ is a retraction depending on the local field F, and every quasi-character χ is of the form $\chi = \tilde{\chi}|\cdot|^s$ for some $s \in \mathbb{C}$ and a ramified character $\tilde{\chi}$ conciding with χ on U.

Proof. It is enough to prove the non-arcimedean case, since we have seen that $\rho(x) = x/\|x\|$ when F is archimedean. Notice that $U = \mathcal{O}^\times$ is a compact subgroup of $F^\times$, then the restriction $\chi|_U$ must be a unitary, and we define $\rho(x) = \varpi^{-v_{\mathfrak{p}}(x)}x$, which is a continous homomorphism with $\rho(\varpi^n u) = u$ for any $n \in \mathbb{Z}$ and $u \in \mathcal{O}^\times$. Hence $c(x) = \chi \cdot (\chi|_U \circ \rho)^{-1}$ defines a quasi-character with $c(u) = 1$ for any $u \in \mathcal{O}^\times$, by lemma 1.5 it shows that c is unramified of the form $c(x) = |x|^s$ for some $s \in \mathbb{C}$, so $\chi = \chi_u|\cdot|^s$ with $\tilde{\chi} = \chi|_U \circ \rho$.

Conversely, we can construct a inverse map by $(c, \chi_s) \mapsto (c \circ \rho) \cdot \chi_s$, which proves the map is an isomorphism. ☐

Remark. The isomorphism is also topological, the retraction ρ induces a map

$$\rho^* : \mathrm{Hom}_{\mathrm{cont}}(U, \mathbb{C}^\times) \to \mathrm{Hom}_{\mathrm{cont}}(F^\times, \mathbb{C}^\times), \quad \chi \mapsto \chi \circ \rho$$

such a map is continous since local field is locally compact and Hausdorff, in this case the map defined in the lemma is a homeomorphism.

Lemma 1.7. Let $\chi : F^\times \to \mathbb{C}^\times$ be a quasi-character of a non-archimedean local field F, then there exists a integer $n \geq 0$ such that χ is trivial on $1 + \mathfrak{p}^n$.

Proof. $\mathbb{S}^1$ is a group has NSS (no small subgroup) properties, then there exists a neighborhood V of 1 in $\mathbb{S}^1$ such that V does not contain any non-trivial subgroup, then $\chi^{-1}(V)$ is a open neighborhood of 1 in $F^\times$. In fact, all open subgroups of $\mathcal{O}^\times$ form a chain as following:

$$\mathcal{O}^\times \supsetneq 1 + \mathfrak{p} \supsetneq 1 + \mathfrak{p}^2 \supsetneq \cdots$$

and form a basis of open neighborhood of 1 in $F^\times$, so there exists a integer $n \geq 0$ such that $1 + \mathfrak{p}^n \subset \chi^{-1}(V)$, which implies that χ is trivial on $1 + \mathfrak{p}^n$. ☐

Hence there is no confusion to take the samllest integer $\mathfrak{f}$ and define it as the **conductor** of quasi-character χ, with a convention that $1 + \mathfrak{p}^0 = \mathcal{O}^\times$, then χ is unramified if and only if its conductor is 0.

Remark. Let $F = \mathbb{Q}_p$ and $\chi : F^\times \to \mathbb{C}^\times$ be ramified character, and then $\chi|_{\mathbb{Z}_p^\times}$ factors through $\mathbb{Z}_p^\times/(1 + p^{\mathfrak{f}}\mathbb{Z}_p)$ with conductor $\mathfrak{f} > 0$ as following

$$\mathbb{Z}_p^\times \longrightarrow \mathbb{Z}_p^\times/(1 + p^{\mathfrak{f}}\mathbb{Z}_p) \xrightarrow{\sim} (\mathbb{Z}/p^n)^\times \xrightarrow{\varphi} \mathbb{S}^1$$

the proof is omitted here, and the interesting result is that the ramified part of χ is related to a primitive dirichlet character φ modulo $p^{\mathfrak{f}}$.

Later we will see that the local zeta integral will be defined on the character group, so it is better to treat it as a complex manifold such that we can talk about the meromorphic continuation and functional equation.

Definition 1.7. Let χ and χ' be two quasi-character of $F^\times$, they are called equivalent if $\chi'|_U = \chi'|_U$, or equivalently, χ/χ' is unramified.

It defines a equivalence relation on $\mathcal{X}(F^\times)$, and the equivalence class of a quasi-character is $[\chi] = \{\chi|\cdot|^s : s \in \mathbb{C}\} = \chi \cdot \mathcal{U}(F^\times)$, and lemma 1.5 shows that $\mathcal{U}(F^\times)$ is indeed a complex lie group of dimension 1, so each class can be identified as a Riemann surface by translation of topological group, and thus $\mathcal{X}(F^\times)$ can be identified as a disjoint union of Riemann surfaces.

Finally, to consider the continuation of local zeta function, Tate introduces a notation to denote the "dual" of a quasi-character as following:

Definition 1.8. Let $\chi \in \mathcal{X}(F^\times)$, its **twisted dual** is defined as $\chi^\vee = \chi^{-1}|\cdot|$.

It is not hard to check that $\chi \mapsto \chi^\vee$ is a homeomorphism on $\mathcal{X}(F^\times)$ with $(\chi^\vee)^\vee = \chi$, and the image of a connected component $[\chi]$ is exactly the component $[\chi^{-1}]$.

1.3 Local Zeta Function

Now we will consider the integrand on $F^\times$, which will be used to define the local zeta integral.

Lemma 1.8. Let dx be the self-dual Haar measure on F defined in the previous, then

(a) $m(E) = \int_E \frac{dx}{|x|}$ defines a Haar measure on multiplicative group $F^\times$;

(b) There exists a unique Haar measure $d^\times x$ on $F^\times$ such that U gets measure 1 and $N(\mathfrak{d})^{-1/2}$ if F is archimedean and non-archimedean respectively, explicitly it is given by

$$d^\times x = \begin{cases} \frac{dx}{|x|} & \text{if } F \text{ is archimedean} \\ \frac{1}{1-N(\mathfrak{p})^{-1}} \cdot \frac{dx}{|x|} & \text{if } F \text{ is non-archimedean} \end{cases}$$

(c) A function f is integrable on $F^\times$ with respect to $d^\times x$ if and only if $x \mapsto f(x)/|x|$ is integrable on $F - 0$ with respect to dx.

Proof. □

Example 1.9. If the number field is $\mathbb{Q}$, then by Ostrowski's theorem, the local field is either $\mathbb{R}$ or $\mathbb{Q}_p$ with p a prime number, so there is no confusion to define an integrand for a proper function f and a complex number s by

$$Z(f, s) = \int_{F^\times} f(x)|x|^s \, d^\times x$$

If $F = \mathbb{Q}$ and we choose $f(x) = e^{-\pi x^2}$, then we can calculate

$$Z(f, s) = 2\int_0^\infty e^{\pi x^2} x^{s-1} \, dx = \pi^{-s/2}\Gamma(s/2)$$

If $F = \mathbb{Q}_p$ and we choose $f(x) = \mathbf{1}_{\mathbb{Z}_p}$, then use the fact that $\mathbb{Z}_p - \{0\} = \bigcup_{n\geq 0} p^n\mathbb{Z}_p^\times$ we can calculate

$$\begin{aligned} Z(f,s) &= \sum_{n\geq 0} \int_{p^n\mathbb{Z}_p^\times} |x|_p^s \, d^\times x \\ &= \sum_{n\geq 0} p^{-ns} \int_{\mathbb{Z}_p^\times} d^\times x \\ &= \sum_{n\geq 0} p^{-ns} = \frac{1}{1-p^{-s}} \end{aligned}$$

Pay attention that in both case, the last equality holds when $\mathrm{Re}(s) > 0$, and by Euler's product formula we can find that the proudct of all local zeta functions is so-called completed zeta function

$$s \mapsto \pi^{-s/2}\Gamma(s/2)\zeta(s)$$

which plays an important role in the functional equation of Riemann zeta function, it can go back to Riemann's famous short paper [1].

In the example above, variable s can be seen as a parameter of unramified quasi-character $\chi_s = |\cdot|^s$, and beyond the example, we should consider other type of zeta function. For instance, a dirichlet L-function asspciated with a dirichlet character χ, which is defined by the following Euler product

$$L(s,\chi) = \prod_{p \text{ prime}} \frac{1}{1-\chi(p)p^{-s}}$$

it suggests that we should consider all type of quasi-character.

Definition 1.9. Fix a function $f \in \mathcal{S}(F)$, the local zeta function associated with f is defined as a integrand on space $\mathcal{X}(F^\times)$ by

$$Z(f,\chi) = \int_{F^\times} f(x)\chi(x) \, d^\times x$$

Let χ be a quasi-character of the class $[\chi_0]$ of the form $\chi_0|\cdot|^s$, it is equivalent to write

$$Z(f,\chi_0,s) = \int_{F^\times} f(x)\chi_0(x)|x|^s d^\times x$$

and we call here s as the complex parameter of χ and $\sigma = \mathrm{Re}(s)$ as the **exponent** of χ.

Review that a function on Riemann surface $f : X \to \mathbb{C}$ is called holomorphic (resp. meomorphically) at $s \in X$ if there exists a local chart (U,ϕ) with $s \in U$ such that $f \circ \phi^{-1}$ is holomorphi (resp. meomorphically) at $\phi(s)$ as a complex function.

Lemma 1.10. Following the notation above, $\chi \mapsto Z(f,\chi)$ defines a holomorphic function on the domain of quasi-character with exponent $\sigma > 0$.

Proof. □

Lemma 1.11 (Tate)**.** Fix any two functions $f, g \in \mathcal{S}(F)$, and let χ be a quasi-character χ of exponent $0 < \sigma < 1$, then $\chi^\vee$ is of exponent $0 < \sigma < 1$ as well, and the following identity holds

$$Z(f,\chi)Z(\hat{g},\chi^\vee) = Z(\hat{f},\chi^\vee)Z(g,\chi)$$

Proof. $\square$

Loosely speaking, the last lemma means the quotient of $Z(f,\chi)$ and $Z(\hat{f},\chi^\vee)$ is independent of the choice of function, so we can definie a meromorphic function on domain of quasi-character of exponent $0<\sigma<1$ by

$$\rho(\chi) = \frac{Z(\hat{f},\chi^\vee)}{Z(f,\chi)}$$

the choice of f should satisfy $\chi \mapsto Z(f,\chi)$ is not zero function, and we call it ρ-factors of local field F. The analytic continuation of ρ-factors will be the key to prove the local functional equation, Tate's strategy is to calculate each case explicitly.

Example 1.12. If $F=\mathbb{R}$, by example 1.4 we have

$$\mathcal{X}(\mathbb{R}^\times) = [\chi_0] \sqcup [\chi_1]$$

with χ_0 the trival character and χ_1 the sign character, and they are representative of even characters and odd characters respectively. In the component $[\chi_0]$, we can choose $f_0(x) = e^{-\pi x^2}$ with $\hat{f}_0 = f_0$; In the compoenent $[\chi_1]$, the choice of f_0 is not proper since its local zeta function will vanish, so we modify it by $f_1(x) = xe^{-\pi x^2}$ with $\hat{f}_1 = i\hat{f}_1$. After some calculation we can conclude the local zeta function

$$Z(f_\epsilon,\chi_{\epsilon,s}) = \pi^{-\frac{s+\epsilon}{2}}\Gamma\left(\frac{s+\epsilon}{2}\right)$$

and its ρ-factor

$$\rho(\chi_{\epsilon,s}) = i^\epsilon 2^{1-s}\pi^s \cos\frac{\pi(s+\epsilon)}{2}\,\Gamma(s)$$

for quasi-character $\chi_{\epsilon,s}(t) = \mathrm{sign}(t)^\epsilon|\cdot|^s$ of exponent $\sigma = \mathrm{Re}(s)\in(0,1)$.

Example 1.13. If $F=\mathbb{C}$, by example 1.4 we have

$$\mathcal{X}(\mathbb{C}^\times) = \bigsqcup_{n\in\mathbb{Z}}[c_n]$$

with $c_n(z) = (z/\|z\|)^n$ the unitary character, and we can choose schwartz functions f_n for each component $[c_n]$ by

$$f_n(z) = \begin{cases} (\bar{z})^n e^{-2\pi\|z\|^2} & n\geq 0 \\ z^{-n}e^{-2\pi\|z\|^2} & n<0 \end{cases}$$

with fourier tranform $\hat{f}_n = i^{|n|}f_{-n}$, and in each case the local zeta function does not vanish, which allows us to calculate the local zeta function

$$Z(f_n,\chi_{n,s}) = (2\pi)^{1-s+\frac{|n|}{2}}\,\Gamma(s+\frac{|n|}{2})$$

and its ρ-factor

$$\rho(\chi_{n,s}) = (-i)^{|n|}(2\pi)^{1-2s}\frac{\Gamma(s+\frac{|n|}{2})}{\Gamma(1-s+\frac{|n|}{2})}$$

for quasi-character $\chi_{n,s} = c_n|\cdot|^s$ of exponent $\sigma \in (0,1)$.

Example 1.14. If F is non-archimedean, let $\mathcal{X}_n(F^\times)$ denote the set of quasi-characters of conductor n, then we have a decomposition

$$\mathcal{X}(F^\times) = \bigsqcup_{n\geq 0} \mathcal{X}_n(F^\times)$$

when $n=0$, $\mathcal{X}_0(F^\times)$ is just all unramified quasi-characters $\mathcal{U}(F^\times)$; when $n=1$, $\mathcal{O}^\times/1+\mathfrak{p}$ can be identified as the multiplicative group of resuidue field $\mathcal{O}/\mathfrak{p}$, so $\mathcal{X}_1(F^\times)$ contains $N(\mathfrak{p})-1$ different classes of quasi-characters; when $n>1$, similar argument shows that $\mathcal{X}_n(F^\times)$ contains $N(\mathfrak{p})^{n-1}(N(\mathfrak{p})-1)^2$ different classes of quasi-characters. Hence the classes of quasi-characters are countable, and we can choose

Lemma 1.15. If F is non-archemedian, then for any

Theorem 1.16 (Tate's Local functional equation)**.**
Let F be a local field and fix a function $f \in \mathcal{S}(F)$, then $\chi \mapsto Z(f,\chi)$ has a meromorphic continuation to whole space $\mathcal{X}(F^\times)$, and there exists a meromorphic function $\rho : \mathcal{X}(F^\times) \to \mathbb{C}$ such that for any quasi-character χ such that , the following functional equation holds

$$Z(\hat{f}, \chi^\vee) = \rho(\chi)Z(f,\chi)$$

where ρ is independent of the choice of f and

Proof. Let ρ as c $\square$

References

[1] B. Riemann, "Ueber die anzahl der primzahlen unter einer gegebenen grosse," *Ges. Math. Werke und Wissenschaftlicher Nachlaß*, vol. 2, no. 145-155, p. 2, 1859.